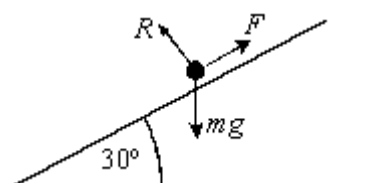


Mark schemes

Q1.

(a)



Correct force diagram

B1

1

(b) $R = 5 \times 9.8 \cos 30^\circ$

Finding normal reaction

M1

$$F = 0.2 + 5 \times 9.8 \cos 30^\circ = 8.5 \text{ N (to sf)}$$

Finding friction

M1

Correct friction

A1

3

(c) $5 \times 9.8 \sin 30^\circ - 0.2 \times 5 \times 9.8 \cos 30^\circ = 5a$

Three term equation of motion

M1

Correct equation

A1

$$a = \frac{5 \times 9.8 \sin 30^\circ - 0.2 \times 5 \times 9.8 \cos 30^\circ}{5}$$

Solving for a

M1

$$= 3.20 \text{ ms}^{-2}$$

Correct a

A1

4

(d) $v^2 = 0^2 + 2 \times 3.203 \times 1.2$

Constant acceleration equation with $u = 0$

M1

Correct equation

A1 ft.

$$v = 2.77 \text{ ms}^{-1}$$

A1 ft.

3

[11]

Q2.

(a) $0 = V \sin \alpha - gt$

Equation for t at max. height

M1

$$t = \frac{V \sin \alpha}{g}$$

Solving for t

M1

Correct t

A1

$$H = V \sin \alpha \times \frac{V \sin \alpha}{g} - \frac{1}{2} \times g \times \left(\frac{V \sin \alpha}{g} \right)^2$$

Substitution into equation for height

M1

Correct expression

A1

$$= \frac{V^2 \sin^2 \alpha}{2g}$$

Correct final result from correct working

A1

6

(b) (i) $6 = \frac{V^2}{2g} \times \frac{3}{4}$

Substitution into given formula

M1

$$V = \sqrt{16g} = 12.5 \text{ ms}^{-1}$$

Solving for V

M1

Correct value

A1

3

$$(ii) \quad t = \frac{2\sqrt{16g} \sin 60^\circ}{g} = 2.213$$

Finding time of flight

M1

Correct time of flight

A1

$$R = \sqrt{16g} \cos 60^\circ \times 2.213 = 13.9 \text{ m}$$

Finding range

M1

Correct range

A1

4

[13]

Q3.

$$(a) \quad 2\mathbf{i} + 14\mathbf{j} = \mathbf{u} + 2(2\mathbf{i} - 3\mathbf{j})$$

Using constant acceleration equation to find $\mathbf{u}$

Correct equation

M1

Correct $\mathbf{u}$ from correct working

A1

$$\mathbf{u} = (2 - 4)\mathbf{i} + (14 + 6)\mathbf{j} = -2\mathbf{i} + 20\mathbf{j}$$

A1

3

$$(b) \quad \mathbf{r} = (-2\mathbf{i} + 20\mathbf{j}) \times 3 + \frac{1}{2} (2\mathbf{i} - 3\mathbf{j}) \times 9$$

Using constant acceleration equation to find $\mathbf{r}$

Correct equation

M1

Correct r

A1

$$3\mathbf{i} + 46.5\mathbf{j}$$

A1

$$r = \sqrt{3^2 + 46.5^2} = 46.6 \text{ ms}^{-1}$$

Finding magnitude

Correct magnitude

M1

5

(c) $\mathbf{v} = (2T - 2)\mathbf{i} + (20 - 3T)\mathbf{j}$

Forming equation for v

M1

$$100 = (2T - 2)^2 + (20 - 3T)^2$$

Forming equation that T must satisfy

M1

$$13T^2 - 128T + 304 = 0$$

Correct Equation

A1

$$T = \frac{128 \pm \sqrt{128^2 - 4 \times 13 \times 304}}{2 \times 13}$$

Solving for T

M1

$$= 4 \text{ or } 5.85$$

Correct solutions

A1

$$T = 4$$

Selecting $T = 4$

A1

6

[14]

Q4.

(a) $120 = \frac{1}{2} a \times 50^2$

Use of constant acceleration equation

M1

Correct equation

A1

$$a = \frac{240}{2500} = 0.096 \text{ ms}^{-2}$$

Correct acceleration from correct working

A1

3

(b) $v^2 = 2 \times 0.096 \times 200$

Use of constant acceleration equation

M1

Correct equation

A1

$$v = 6.20 \text{ ms}^{-1}$$

Correct velocity

A1

3

(c) $200 = \frac{1}{2} \times 0.096 t^2$

Use of a constant acceleration equation

M1

Correct equation

A1

$$t = \sqrt{\frac{400}{0.096}} = 64.5 \text{ s}$$

Correct time

A1

3

(d) $F - 250 \times 9.8 = 250 \times 0.096$

Three term equation of motion

M1

Correct equation

A1

$$F = 2470 \text{ N (to 3sf)}$$

Correct force

A1

3

[12]

Q5.

$$(a) \quad \mathbf{r} = (4\mathbf{i} + 7\mathbf{j})t + \frac{1}{2}(-2\mathbf{i} + 4\mathbf{j})t^2$$

Finding r

M1

$$= (4t - t^2)\mathbf{i} + (7t - 2t^2)\mathbf{j}$$

Correct r

A1

2

$$(b) \quad 4t - t^2 = 3$$

Forming equation for one component

M1

$$t^2 - 4t + 3 = 0$$

$$(t - 1)(t - 3)$$

Solving quadratic equation

m1

$$(t - 1)(t - 3)$$

Solving quadratic equation

m1

$$t = 1 \text{ or } t = 3$$

Two solutions

A1

$$7t - 2t^2 = 5$$

Forming and solving quadratic equation

M1

$$2t^2 - 7t + 5 = 0$$

$$(2t - 5)(t - 1)$$

$$t = 1 \text{ or } t = 2.5$$

Two solutions

A1

$$t = 1$$

Selecting correct solution

A1

6

[8]

Q6.

(a) $15^2 = 20^2 + 500a$

Use of constant acceleration equation

M1

Correct equation

A1

$$a = -0.35 \text{ ms}^{-2}$$

Correct acceleration from correct working

A1

3

(b) $P - 1200 \times 9.8 \cos 86^\circ = 1200 \times (-0.35)$

Three term equation of motion

M1

Correct equation

A1

Solving for P

m1

$$P = 400 \text{ N}$$

Correct P

A1

4

(c) $P = 400 + 300 = 700 \text{ N}$

Including resistance force

	Correct value	M1	
		A1	2
(d)	Expect air resistance force to decrease as car moves up the slope as it is slowing down		
	Air resistance		
	Describing variation		

B1
2

[11]

Q7.

(a) $s = \frac{1}{2} \times 2 \times 4^2 = 16 \text{ m}$

Using $s = ut + \frac{1}{2} at^2$ with $u = 0$ oe
cao

M1 A1
2

(b) $v = 2 \times 4 = 8 \text{ ms}^{-1}$

Using $v = u + at$ with $u = 0$ oe
cao

M1 A1
2

(c) $100 - 16 = 8t$

Using 100 minus answer to part (a)

M1

$$t = \frac{100 - 16}{8} = 10.5 \text{ s}$$

Correct equation
10.5 cao

A1 A1

Total Time = $10.5 + 4 = 14.5\text{s}$

Adding 4 to get 14.5

A1ft
4

[8]

Q8.

(a) (i) $60 \times 9.8 - T = 60a$

Equation of motion for one boy
Correct equation

M1 A1

$$T - 40 \times 9.8 = 40a$$

Equation of motion for other boy
Correct equation

M1

$$588 - (40a + 392) = 60a$$

$$a = 1.96 \text{ ms}^{-2}$$

ag Correct answer from correct working

A1

5

(ii) $T = 40 \times 1.96 + 392 = 470 \text{ N (to 3 sf)}$

Substituting for a and solving for T
awrt 470

M1 A1 M1

2

(b) $v^2 = 0^2 + 2 \times 1.96 \times 2 = 7.84$

Finding v^2 using $s = 2$, $u = 0$ and $a = 1.96$
Correct v or v^2

M1 A1

$$0^2 = 7.84 + 2 \times (-9.8)s$$

Equation for s with $v = 0$ and $a = \pm g$

m1 A1

$$s = 0.4$$

Solving for s

m1

$$h_{\max} = 2 + 0.4 = 2.4 \text{ m}$$

Solving for s
Obtaining 0.4
Adding 2 to get 2.4

m1 A1 A1

7

Q9.

(a) $0 = V \sin \alpha t - gt^2$

Equation for height equal to zero

B1

$$t = 0 \text{ or } t = \frac{2V \sin \alpha}{g}$$

Finding t

Correct time of flight

M1 A1

$$\text{Range} = V \cos \alpha \times \frac{2V \sin \alpha}{g}$$

Expression for horizontal displacement

Substituting the time of flight

M1 M1

$$= \frac{V^2 \sin 2\alpha}{g}$$

Correct final answer from correct working

A1

6

- (b) Because the maximum value of $\sin 2\alpha$ is 1.

Reason that uses maximum of sin being 1

B1

$$\alpha = 45^\circ$$

$$45^\circ$$

B1

2

(c) (i) $\frac{V^2}{g} = 35$

Substituting 35

M1

$$V = \sqrt{35g} = 18.5 \text{ m s}^{-1}$$

cao

A1

2

$$(ii) \quad R = \frac{18.5^2 \sin 60^\circ}{g} = 30 \text{ m}$$

Substituting $\alpha = 30^\circ$
30 m

M1 A1

2

(or $R = 35 \sin 60^\circ = 30$)

[12]

Q10.

(a) (i) $\vec{AB} = (70\mathbf{i} - 40\mathbf{j}) - (30\mathbf{i} - 35\mathbf{j}) = 40\mathbf{i} - 5\mathbf{j}$

cao

B1

1

(ii) $40\mathbf{i} - 5\mathbf{j} = 10\mathbf{v} + \frac{1}{2} (0.2\mathbf{i} + 0.1\mathbf{j}) \times 100$

Using their $\vec{AB}$
Correct RHS of the equation

M1 A1

$10\mathbf{v} = 30\mathbf{i} - 10\mathbf{j}$
 correct $\mathbf{v}$
 $\mathbf{v} = 3\mathbf{i} - \mathbf{j}$

A1

3

(b) $3\mathbf{i} - \mathbf{j} = \mathbf{u} + (0.2\mathbf{i} + 0.1\mathbf{j}) \times 10$

Correct RHS

M1 A1

$\mathbf{u} = \mathbf{i} - 2\mathbf{j}$
 Correct $\mathbf{u}$

A1

3

(c) $30\mathbf{i} - 35\mathbf{j} = (\mathbf{i} - 2\mathbf{j}) \times 10 + \frac{1}{2} (0.2\mathbf{i} + 0.1\mathbf{j}) \times 100 + \mathbf{r}_0$
Equation of correct form for $t = 10$ or 20 with their $\mathbf{u}$
Correct equation with their $\mathbf{u}$

M1 A1

$$30\mathbf{i} - 35\mathbf{j} = 20\mathbf{j} - 15\mathbf{j} + \mathbf{r}_0$$

$$\mathbf{r}_0 = 10\mathbf{i} - 20\mathbf{j}$$

Correct position vector

A1

3

[10]

Q11.

(a) $34 = 8 + 20a$

Using constant acceleration equation

Correct equation

M1 A1

$$a = 1.3 \text{ ms}^{-2}$$

Correct a

A1

3

(b) $s = \frac{1}{2} \times (8 + 34) \times 20 = 420\text{m}$

Using constant acceleration equation

Correct distance

A1

2

(c) $F - 400 = 1250 \times 1.3$

Three term equation of motion

Correct equation

M1 A1

$$F = 2025 \text{ N}$$

Correct force

A1

3

[8]

Q12.

(a) $\mathbf{F} = (2 + 6 - 2)\mathbf{i} + (3 - 8 + 10)\mathbf{j}$
Summing the forces

M1

$= 6\mathbf{i} + 5\mathbf{j}$
Correct resultant

A1

2

(b) $\mathbf{a} = \frac{1}{8} (6\mathbf{i} + 5\mathbf{j}) = 0.75\mathbf{i} + 0.625\mathbf{j}$
Dividing by 8 to get correct value

B1

1

(c) (i) $\mathbf{v} = (3\mathbf{i} - 2\mathbf{j}) + 40(0.75\mathbf{i} + 0.625\mathbf{j})$
use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1

$= 33\mathbf{i} + 23\mathbf{j}$
Correct $\mathbf{v}$

A1

2

(ii) $\mathbf{r} = (3\mathbf{i} - 2\mathbf{j}) \times 40 + \frac{1}{2} (0.75\mathbf{i} + 0.625\mathbf{j}) \times 40^2$
Vector constant acceleration equation to find $\mathbf{r}$
Correct equation

M1 A1

$= 720\mathbf{i} + 420\mathbf{j}$
Correct position vector

A1

$r = \sqrt{720^2 + 420^2} = 834 \text{ m}$

Finding magnitude
Correct magnitude

M1 A1

5

[10]

Q13.

(a) $5^2 = 1^2 + 2 \times 4a$

Using constant acceleration equation to find a

M1 A1

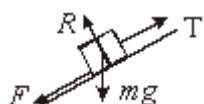
$a = 3 \text{ ms}^{-2}$

Correct equation

A1

3

(b) (i)



Correct force diagram

B1

1

(ii) $R = 20 \times 9.8 \cos 30^\circ = 170 \text{ N}$

Finding the normal reaction

M1

$F = 0.4 \times 20 \times 9.8 \cos 30^\circ = 67.9 \text{ N}$

Using $F = \mu R$

M1 A1

3

(iii) $T - 67.9 - 20 \times 9.8 \sin 30^\circ = 20 \times 3$

Four term equation of motion

Correct equation

M1 A1

$T = 226 \text{ N}$

Solving for T

Correct T

m1 A1

4

[11]

Q14.

(a) $20^2 = 12^2 + 2 \times a \times 200$

*Using a constant acceleration equation
with a unknown*

M1

$$a = \frac{400 - 144}{400} = 0.64 \text{ ms}^{-2}$$

Correct equation

Correct result from correct working

A1/A1

3

(b) $20 = 12 + 0.64t$

*Using a constant acceleration equation
with t unknown*

M1

$$t = \frac{20 - 12}{0.64} = 12.5 \text{ seconds}$$

Correct equation

Correct result from correct working

A1/A1

3

(c) (i) $F = 1200 \times 0.64 = 768 \text{ N}$

Correct answer only

B1

1

(ii) $F - 450 = 768$

Three term equation of motion

M1

$$F = 1220 \text{ N}$$

Correct result

Accept 1218

A1

2

[9]

Q15.

(a) $6060 - 600 \times 9.8 = 600a$

Three term equation of motion

M1

$$a = \frac{6060 - 5880}{600} = 0.3 \text{ ms}^{-2}$$

Correct acceleration from correct working

A1

2

(b) $8 = \frac{1}{2} \times 0.3t^2$

*Using a constant acceleration equation
with t unknown*

M1

$$t = \sqrt{\frac{16}{0.3}} = 7.30 \text{ seconds}$$

Correct equation

Correct result

Accept 7.3

A1/A1

3

(c) $v^2 = 2 \times 0.3 \times 8$

*Using a constant acceleration equation
with v unknown and u = 0*

M1

$$v = \sqrt{4.8} = 2.19 \text{ ms}^{-1}$$

Correct velocity

A1

2

[7]

Q16.

(a) $-25 = 40\sin 5^\circ t - 4.9t^2$

M1/A1/A1

$$4.9t^2 - 40\sin 5^\circ t - 25 = 0$$

Equation for vertical motion

LHS correct

RHS correct

$$t = \frac{40 \sin 5^\circ \pm \sqrt{(40 \sin 5^\circ)^2 - 4 \times 4.9 \times (-25)}}{2 \times 4.9}$$

Solving quadratic equation
Correct method

m1/A1

$$= 2.642 \text{ or } -1.931$$

Hits water after 2.64 seconds

Selecting 2.64

A1

6

(b) $x = 40 \cos 5^\circ \times 2.642 = 105 \text{ m}$

Equation for horizontal motion
Correct range

M1/A1

2

(c) $v_x = 40 \cos 5^\circ$

$$v_y = 40 \sin 5^\circ - 9.8 \times 2.642$$

Horizontal component at impact
Vertical component at impact
Vertical component correct

M1/M1/A1

$$v = \sqrt{(40 \cos 5^\circ)^2 + (40 \sin 5^\circ - 9.8 \times 2.642)^2}$$

Finding magnitude

M1

$$= 45.7 \text{ ms}^{-1}$$

Correct final answer

A1

5

[13]

Q17.

(a) $T - 0.8 \times 5 \times 9.8 = 5a$

Equation of motion for one particle

M1

$$T = 5a + 39.2$$

Correct equation

A1

$$3 \times 9.8 - T = 3a$$

Equation of motion for other particle

M1

$$29.4 - (5a + 39.2) = 3a$$

Correct equation

A1

$$-9.8 = 8a$$

Solving for a

M1

$$a = -1.225 \text{ ms}^{-2}$$

Correct a

A1

6

(b) (ii) $T = 5a + 39.2$

Substituting and solving for T

M1

$$T = 33.1 \text{ N}$$

Correct T

A1

2

(c) $0^2 = 2^2 + 2 \times (-1.225)s$

Use of CA equation with $v = 0$

Correct equation (allow $a = 1.225$)

M1/A1

$$s = \frac{4}{2.45} = 1.63 \text{ m}$$

*Correct answer from correct working
(allow 1.64)*

A1

3

[11]

Q18.

(a) $44\mathbf{i} + 28\mathbf{j} = 10(4\mathbf{i} + 2\mathbf{j}) + 50\mathbf{a}$

Equation to find a using position vector

Correct equation

M1/A1

$\mathbf{a} = 0.08\mathbf{i} + 0.16\mathbf{j}$

Solving for a

Correct a

M1/A1

4

(b) $\mathbf{r} = (4\mathbf{i} + 2\mathbf{j})t + \frac{1}{2}(0.08\mathbf{i} + 0.16\mathbf{j})t^2$

Equation for r

One component correct

M1/A1

$= (4t + 0.04t^2)\mathbf{i} + (2t + 0.08t^2)\mathbf{j}$

Second component correct

A1

3

(c) (i) $96 = 4T + 0.04T^2 \quad 72 = 2T + 0.08T^2$

Forms and solves equation for one component

M1

$T = 20 \text{ or } -120 \quad T = 20 \text{ or } -45$

Correct solutions

A1

Hence $T = 20$

Forms and solves equation for other component

Correct solutions plus correct conclusion

M1/A1

4

(d) $\mathbf{v} = (4\mathbf{i} + 2\mathbf{j}) + (0.08\mathbf{i} + 0.16\mathbf{j}) \times 20$

Finding v at t = 20

M1

$= 5.6\mathbf{i} + 5.2\mathbf{j}$

Correct v

A1

$$v = \sqrt{5.6^2 + 5.2^2} = 7.64 \text{ ms}^{-1}$$

Finding correct speed

A1

3

[14]



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